

Team Members: Abigail Chase (abechase@ucsc.edu) (eluciaaa), Diego Barajas Mosqueda (divbaraj@ucsc.edu) (yooholup), Vivian Kim (vekim@ucsc.edu) (vekim00100)

Final Project

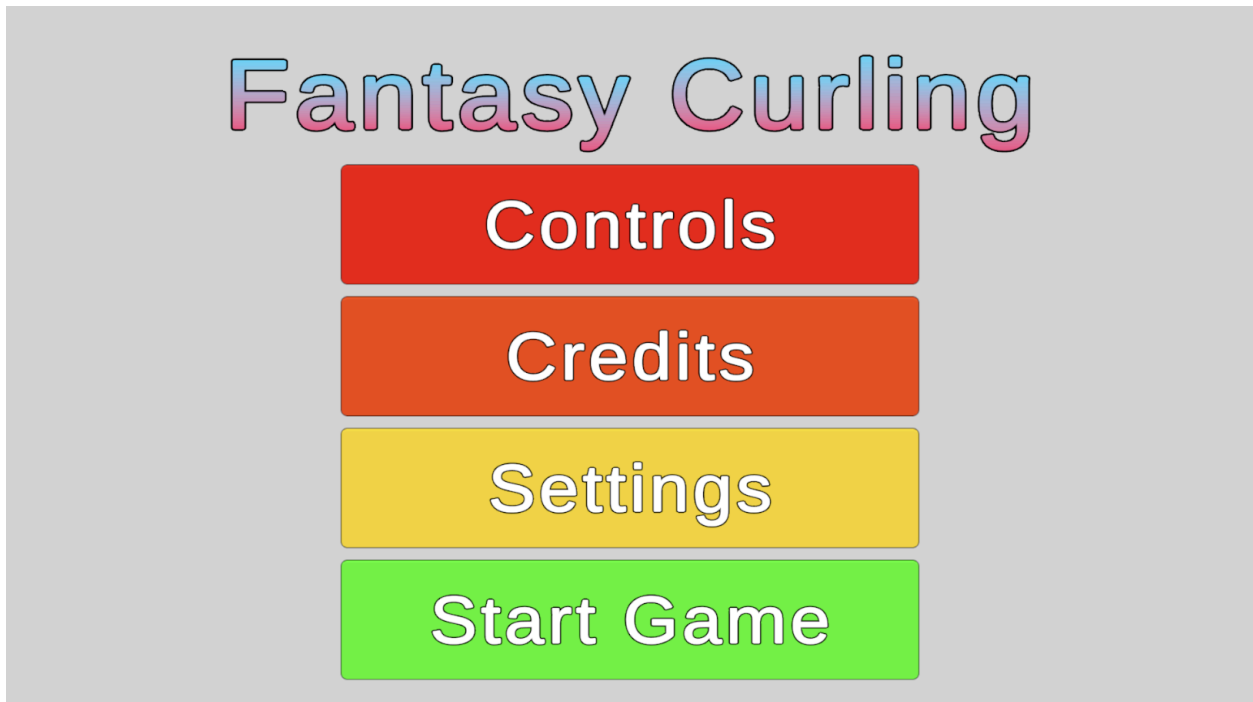
We chose to expand Assignment 2 (Curling) for this assignment. We added 3 new levels, all with their own individual aesthetic theming. The first level takes place on a frozen lake in a snowy mountain area, the second level takes place on a slab of obsidian in a volcano, and the third level takes place up in the clouds. We have a function to change the ice material for each map.

We also added difficulty scaling and individual hazards to each level. We added different ice physics per level. For the hazards, we added fragile ice areas for the frozen lake, which apply a small slow if the rock collides with any of them. We added lava pools for the volcano, which apply a much bigger slow when collided with. We added strong winds for the clouds, which apply a directional push and slow to the rock when collided with. Through these hazards and ice physics, we were able to really apply the difficulty scaling we were looking for.

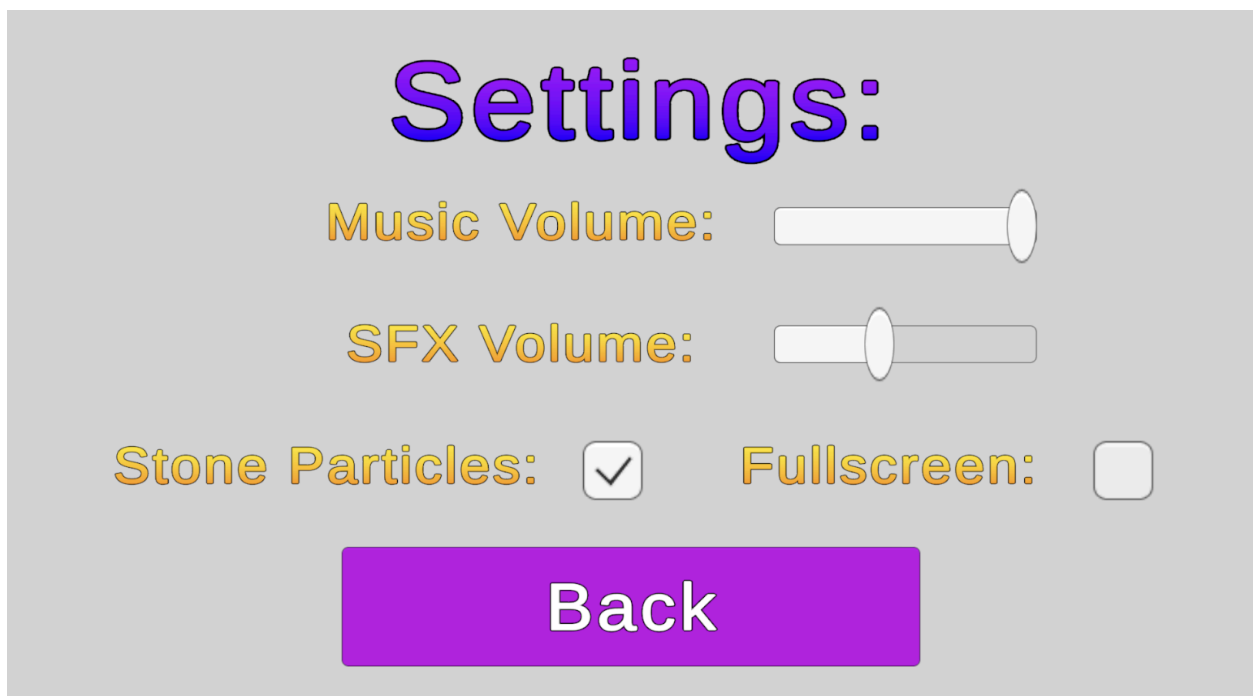
Finally, we added some vfx, audio, and settings. We added ice scraping particles and a thin ice trail coming out the sides of the rock, like in real curling. We have a function to change the color of these particles/trails depending on the map. We also added background music for each individual level, along with SFX for rock sliding (with a function to decrease the volume as the rock slows down), rocks hitting each other, round end, and victory. We added a main menu scene with controls, settings, credits, etc. The settings have controls for music volume (slider),

sfx volume (slider), stone particles (toggle) and fullscreen (toggle). The volume and fullscreen settings save player preference.

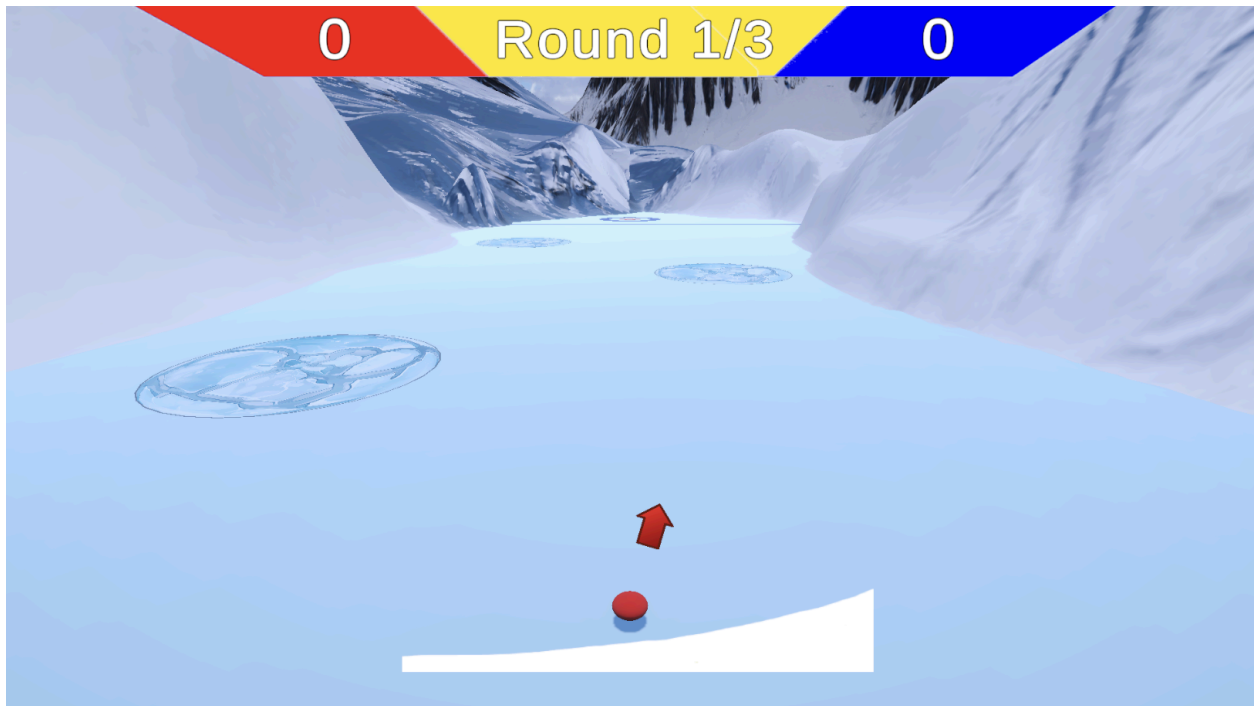
Main Menu



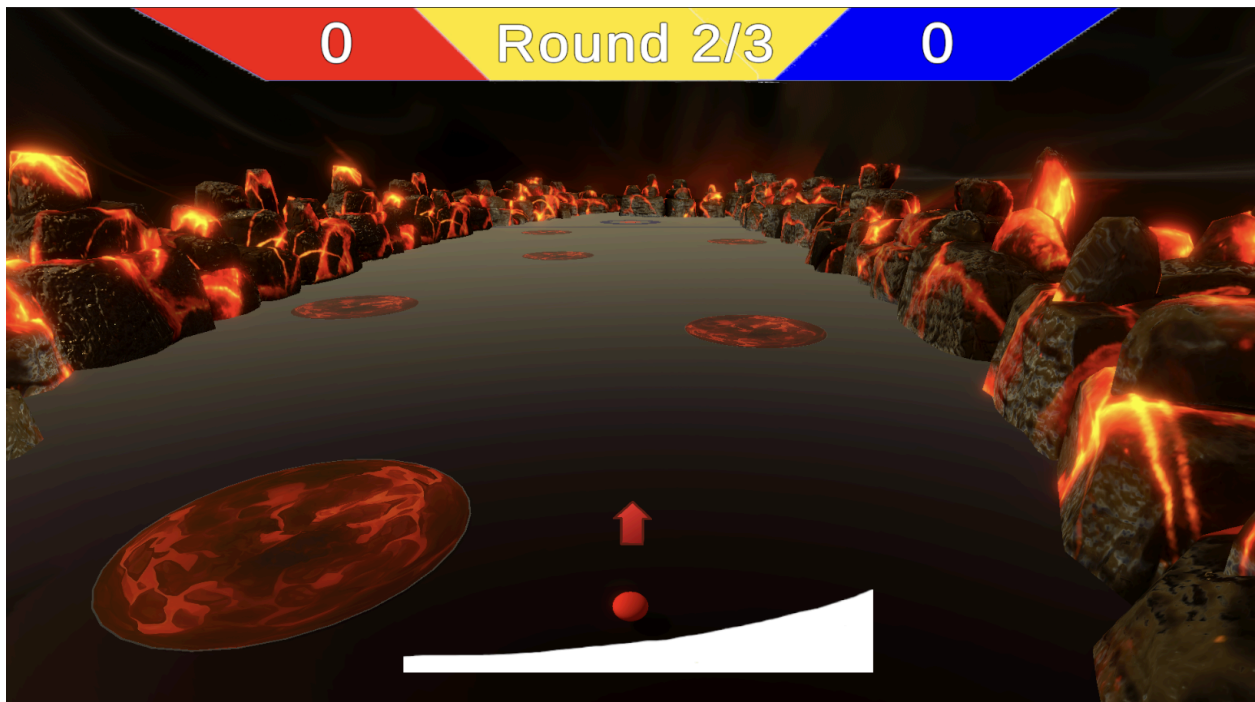
Settings Panel



Fragile Ice Hazards on the Frozen Lake Map



Lava Pool Hazards on the Volcano Map



Strong Winds Hazards on the Clouds Map



Audio Assets Used (also listed in the credits section of the main menu):

Frozen Lake BG Music: Melting Ice (pixabay: Royalty Free Music) Creative Commons 0

Volcano BG Music: dirty grind.aif by griffon_designs License: Attribution 4.0

Clouds BG Music: The Leading Edge (orchestral climax anthem) by SondreDrakensson License: Attribution 4.0

Stone Sliding SFX: Curling: stone slides on ice by Quincy Graham on [DeadSounds.com](https://deadsounds.com) Creative Commons 0

Stone Hitting SFX: Curling: a rock hitting another rock by Brandon Lockman on [DeadSounds.com](https://deadsounds.com) Creative Commons 0

Round End SFX: Gaming Victory (pixabay: EAGLAXLE) Creative Commons 0

Victory SFX: 11L-Victory-1749704552668 (pixabay: AstralSynthesizer) Creative Commons 0

Abigail Chase Reflection:

Contributions:

- Created all 3 maps
- Created all particles and trails
- Create all 3 ice physics presets
- Added all hazards and effects
- Created code to swap between maps and adjust ice color/particle color and ice physics depending on the map
- Added hazard text UI
- Added main menu scene, buttons and their functions, and UI
- Added background music
- Added SFX

Reflection:

During this project, I learned how to improve a physics-based game by adding new mechanics, levels, and visual effects. I worked with Rigidbody forces, collision detection, triggers, and particle systems to create features like wind hazards and different ice conditions. I also learned how important it is to make sure gameplay and visuals work together. I also learned how to level design. It was really tough to figure out how to make all 3 levels look good but also function as maps (blocking the player from going out of bounds and not feeling too bad to curl on). Balancing those two aspects was definitely the hardest part of this assignment, as I had to keep playtesting and adjusting terrain/props to feel good and look good.

Throughout the quarter I learned how to properly do all the main parts of a polished game: physics, aesthetics, post processing, and UI. For the final project, I combined all this knowledge to make a game I can safely say I'm proud of. If I had more time, I would definitely add more levels, maybe some enemy AI, and some dynamic sound effects (based on where the rock lands), ramping up music based on if many rocks are close to the center, etc. However, I feel everything that was put in the game was done well, so I wouldn't change any existing things. Overall, this project helped me improve my Unity skills and understand how to turn a basic prototype into a more complete game.

Diego Barajas Mosqueda Reflection:

Contributions:

For this project, I focused on adding post-processing effects to each of the different level types in our game, adding appropriate values for the three maps to the scripts that change the environment. I also changed the position of the directional light in each level type to give off the effect of differing times of day for each level type. Since this work was very similar to that of Assignment 3, I didn't personally run into any technical issues while working on this assignment.

Reflection:

Throughout the quarter, I learned a lot about both creating games in Unity as well as about myself. This quarter was the first time I had ever worked with Unity, both in this class and in CMPM 121, which focused on a 2D game rather than the 3D environments we utilized in this course. I learned a lot about both the physics of 3D games as well as some of the graphical work that goes into creating 3D games that look pleasing on top of playing nicely. I also realized from

Assignment 3 especially that I prefer working with graphics and post processing settings above much of anything else in game design, so in the future I'd like to take more courses that focus on this particular aspect of game design. If I had more time, I think I would have attended more lectures to learn even more about the different aspects of game development, as well as further expansion of the aspects of game development that I know I enjoy.

Vivian Kim Reflection:**Contribution:**

For the project, my intended focus areas for the project were the main menu system and audio integration. During development, those components progressed quickly and were implemented as part of the team's broader effort. I remained involved in reviewing the game's functionality, testing features as they were integrated, and helping ensure the overall experience was cohesive and polished.

Reflection:

Throughout the quarter, I learned how game development extends beyond individual feature implementation. Successful projects depend on planning, testing, iteration, and collaboration between team members. Given more time, I would take a more proactive role in implementing additional aesthetics and themes for the game, as my interests lie more on the artistic and visual design side of game development. Overall, this project helped me gain a greater appreciation for the importance of teamwork and adaptability throughout the development process.